PULSE Introduction

Document Overview

This document introduces PULSE as a cross-platform wellbeing application and explains its purpose, intended users, principal features, operating rules, cross-feature workflows, and implementation technology. It is written for the stakeholder, assessors, future maintainers, and project contributors who need a concise guide to what the current application does and how its major features work together.

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Reading Guide

Sections 1 to 3 explain why PULSE exists, what is included in the current project, and who it is designed for. Section 4 describes each feature through a function description, rules, workflows, and examples, including the connections between scoring, missions, trackers, finance, savings, social sharing, analytics, and tutorials. Section 5 lists the application technologies, exact project versions, architecture, testing tools, and official development or download websites. The document describes the current implementation; future ideas are included only when they are explicitly labelled as recommendations.

1 Project Overview

PULSE is a cross-platform mobile wellbeing application designed to help people understand and improve the balance of their everyday lives. It translates the Balanced Life model into five connected domains: Spirituality, Family & Friends, Health & Fitness, Finance, and Productivity. The app combines guided onboarding, entry quiz, daily mission engines, journals, physical activity tracker, budget tracker, saving plan, progress insights, and social encouragement in one place.

PULSE establishes a baseline through an entry assessment, then presents individual domain scores together with an overall Balanced Life Score. As users record meaningful actions and reflections over time, the app updates their progress and recommends small, achievable next steps. This makes wellbeing an ongoing, practical process rather than a one-off assessment.

2 Purpose and Scope

The purpose of PULSE is to give users one clear and approachable system for building sustainable balance across mental wellbeing, relationships, physical health, financial habits, and work or study. It is intended to reduce the fragmentation created by using separate apps for habits, budgeting, exercise, productivity, journaling, and social support.

The project scope covers account creation and profile management, an entry assessment, domain and overall scoring, daily missions, domain-specific tracking tools, progress history and analytics, social interaction, privacy controls, and guided first-use tutorials. The current application is designed as a cross-platform mobile experience, with a shared codebase that can also support the web.

3 Target Users

PULSE is primarily intended for goal-oriented people who want more structure, self-awareness, and balance in their lives. Stakeholder discussions identified a core audience ranging from university students to working adults around 40 years old, while the broader design also supports professionals experiencing burnout, first-time parents, athletes, and other users managing competing priorities.

Because the audience has varied technical confidence and different daily routines, the interface emphasises simple navigation, clear explanations, manageable actions, and visible progress. The design is intended to help both new and experienced users engage without requiring specialist knowledge of wellbeing, fitness, or personal finance.

4 Main Features

Each feature is documented separately as Function Description, Rules, and Workflows and Examples. The Function Description also explains how the feature exchanges data with, affects, or triggers behaviour in other parts of PULSE, making the end-to-end user journey explicit.

4.1 Account, Authentication and Privacy

Function Description

Account, Authentication and Privacy establishes the signed-in identity used to separate every user's personal records, including quiz baselines, scores, missions, journals, workouts, finance data and tasks. The same display name and profile image are used in friend search, accepted-friend posts and one-to-one messages. Authentication and consent also control the next screen: a new user proceeds to privacy consent and the Entry Quiz, while a returning eligible user proceeds to the main app; required consent gates the sensitive Health and Spirituality features.

Rules

- The current app supports email-and-password accounts. Full name, a valid email address and a password of at least six characters are required.
- The user must verify the registration email before signing in. Google and Apple sign-in buttons are placeholders and are not active authentication methods.
- The privacy notice must be acknowledged before entering the app. Consent for health and spiritual information is required because those features process sensitive information; analytics and reminders are optional.
- Changing a password requires the current password and a new password of at least six characters.

- Authentication persists between sessions until the user signs out.

Workflows and Examples

1. Create and verify an account: Open Sign Up, enter full name, email and a six-or-more-character password, submit, then open the verification email before signing in. Example: Taylor enters taylor@example.com and a valid password, receives the Firebase verification email, verifies it, and then signs in.
2. Sign in or recover access: Enter the verified email and password. If the password is forgotten, use Forgot Password and enter the registered email. Example: Taylor requests a reset link for taylor@example.com and follows the email instructions to set a new password.
3. Complete privacy setup: Read the notice, choose whether to allow optional analytics and reminders, and continue to consent to the sensitive-data features. Example: Taylor allows reminders but leaves analytics disabled; the required health-and-spiritual consent is still recorded.
4. Update the profile: Open Profile, choose Edit profile, change the display name or profile photo, and save. Example: Taylor changes the display name to Taylor R. and selects a new avatar from the device.
5. Change the password: In Edit profile, enter both the current password and a new password with at least six characters. Example: Taylor enters the existing password, enters a stronger new password, and taps Update Password.

4.2 Entry Quiz

Function Description

The Entry Quiz collects one self-rating for each of the five Balanced Life domains and converts each 1-to-5 answer into an initial 20-to-100 baseline. These baseline values seed the domain cards and Balanced Life Score on Home, become the starting reference for weekly score calculations, and influence which lower-scoring domains Daily Missions prioritises. As later activity changes the scores, weekly snapshots built from the baseline and new evidence are passed to Progress Analytics.

Rules

- The quiz always includes Spirituality, Family & Friends, Work / Productivity, Health, and Financial Wellbeing.
- Each domain uses a five-point scale from Never (1) to All the time (5).
- All five questions are mandatory; submission remains disabled until every domain has an answer.
- Each answer is multiplied by 20, so the initial domain baseline ranges from 20 to 100.

Workflows and Examples

1. Complete the initial assessment: Select one response for every domain and tap Submit check-in. Example: Taylor chooses 3 for Spirituality, 4 for Relationships, 2 for Productivity, 4 for Health and 3 for Finance; the starting scores become 60, 80, 40, 80 and 60.

4.3 Domain Scores and Balanced Life Score

Function Description

The scoring system combines the Entry Quiz or current week-start score with this week's reflections, action points and active days to calculate five domain scores and one Balanced Life Score. Mood and journal check-ins, saved workouts, first-time task completions, logged expenses, domain actions and completed Daily Missions supply scoring evidence. The resulting scores are shown on Home, used to choose Daily Mission focus domains, and saved as weekly snapshots for Progress Analytics.

Rules

- Each domain calculation combines reflection (30%), action points (40%) and consistency across active days (30%).
- Reflection values use the 1-to-5 scale and are converted to 20-to-100 scores. Action scores are measured against a weekly target and capped at 100. Consistency equals active days divided by seven.
- Weekly targets are 200 action points for Spirituality, Relationships and Finance, 250 for Productivity, and 300 for Health.
- Typical rewards are 50 points for a journal, mindfulness activity, relationship action, exercise or saving/budget action, and 25 points for a completed task or logged expense.
- At the beginning of each Monday-to-Sunday week, the app freezes the starting domain scores. The week's observed performance maps to an adjustment from -2 to +2 points, with final scores capped between 0 and 100.
- No activity means the previous score is carried forward. A reflection without an action updates reflection data but does not by itself raise or lower the domain score.
- The Balanced Life Score is 50% the average of the five domain scores and 50% the evenness of those scores.

Workflows and Examples

1. Establish the baseline: Complete the Entry Quiz so every domain has a starting score. Example: A Health response of 4 creates an initial Health score of 80.
2. Add scoring evidence through normal features: Record activities, tasks, expenses, savings, journals or completed missions; the relevant feature logs the matching action automatically. Example: Logging a run adds a 50-point Health exercise action without the user manually entering score points.

3. Build consistency across the week: Use a feature on multiple days rather than recording everything on one day. Example: Health actions on Monday, Wednesday and Saturday produce three active Health days for the consistency component.
4. Review the result: Return to Home or Progress Analytics after the score refreshes. Example: Taylor compares the five domain scores and sees that a more even set of scores can improve the overall Balanced Life Score even if the average changes only slightly.

4.4 Daily Missions

Function Description

Daily Missions reads the current five domain scores and can keep the clearly lowest and second-lowest domains in the week's daily mix before filling the remaining slots from other domains. Completing a mission writes the matching action to that domain's scoring record, so later score recalculation can update Home and Progress Analytics. Mission results are also archived into Mission History, and completing all three can open a prompt to share the achievement in the Social feed.

Rules

- Exactly three missions are generated for each calendar day, and the three missions come from three different domains.
- At the start of each Monday-to-Sunday week, domains are ordered by their current scores. The lowest domain becomes a fixed daily focus only when it is strictly lower than the second-lowest. The second-lowest also becomes fixed only when it is strictly lower than the third-lowest. Ties do not create an additional fixed focus.
- Any remaining slots are selected randomly from the other domains. The selected weekly focus remains in place until the following Monday-to-Sunday week.
- The app prefers mission text that has not already appeared during the same week. A domain's full mission pool is reused only after its unused options are exhausted.
- The same mission set and completion state are restored whenever the user returns on the same day. A new day archives the previous set before generating the next set.
- Each transition from incomplete to complete logs the corresponding domain action. Changing a mission back to incomplete updates the checklist but does not remove an action already recorded for scoring.
- Mission history is stored by date and summarises weekly progress, completion by domain and all-time completion.

Workflows and Examples

1. Review today's missions: Open Home or the full Daily Missions screen and read the three assigned actions and their domain labels. Example: Taylor receives Track your spending for today, Move your body, and Message someone you care about.

2. Complete a mission: Perform the activity and tap its checkbox. Example: After recording the day's spending, Taylor checks the Finance mission; it becomes completed and a Finance action is logged.
3. Undo a checklist state: Tap a completed mission again when it was marked accidentally. Example: Taylor unchecks Move your body after realising the workout has not happened yet; the checklist changes back to incomplete, while any action already written to scoring is not deleted.
4. Review mission history: Open History from the Daily Missions screen and inspect daily records and domain totals. Example: Taylor checks the current week and sees five of seven assigned Health missions completed.

4.5 Mood Tracker and Journal

Function Description

Mood Tracker and Journal stores the user's daily mood, emotion tags and optional private reflection. Saving a check-in records a Spirituality reflection and journal action, so the Spirituality score can change on Home, affect Daily Mission priorities and appear in later Progress Analytics snapshots. After saving, PULSE may offer an achievement-style Social share, but the private journal text is not posted unless the user explicitly chooses to share the generated content.

Rules

- If today has no saved mood, the Spirituality screen offers a daily check-in. The user may select one of five mood levels or skip.
- A mood must be selected before continuing to the journal step. Journal text and emotion tags are optional.
- One displayed mood-and-journal record is stored per local calendar day for each user on that device. Starting another check-in on the same day replaces that day's displayed record.
- Saving a mood records a Spirituality reflection. Saving non-empty journal text also records a Spirituality journal action.
- The dashboard shows the latest seven-day mood pattern and up to four recent journal entries. Past entries are view-only in the current interface.

Workflows and Examples

1. Record a mood only: Open Spirituality, choose a mood, continue, and select Continue without writing. Example: Taylor chooses Good and skips the text field; the mood is saved for today.
2. Record a mood with a journal: Choose a mood, select any matching emotion tags, write a reflection, and save. Example: Taylor chooses Okay, adds Tired and Hopeful, writes Took a short walk and felt calmer, then saves.

3. Skip the check-in: Choose Skip for now when no entry should be created. Example: Taylor opens Spirituality during a meeting, skips the prompt, and returns later to record the mood.
4. Update today's entry: Tap Log today's mood again and complete the check-in. Example: Taylor changes the day's mood from Okay to Good and writes a new reflection; the displayed record for today is replaced.
5. Review patterns: Use the seven-day chart and recent entries below it. Example: Taylor notices lower moods on several workdays and opens the most recent journal entries for context.

4.6 Physical Activity Tracker

Function Description

Physical Activity Tracker records cardio and strength workouts and lets saved activities be reused as templates. Saving a real workout records a Health exercise action, so the Health score can update on Home, influence Daily Mission selection and flow into Progress Analytics; simply saving or opening a template does not create scoring evidence until the workout itself is logged. A completed activity can also open a Social sharing prompt.

Rules

- Every activity requires a name and an activity date. Description is optional.
- A Cardio activity requires both a distance greater than zero and a duration greater than zero.
- A Strength activity requires at least one named exercise. Every set must contain a positive weight, a positive whole-number repetition count, and a positive whole-number set count.
- Saving a new valid activity logs one 50-point Health exercise action. Editing an existing record does not create another exercise reward.
- Activities can be edited or permanently deleted after confirmation.
- Any saved activity can be marked as a template. Using a template fills the New Activity form but does not save a new workout until the user reviews and submits it.

Workflows and Examples

1. Log a Cardio activity: Open Features, choose Physical Activity, stay on New, select Cardio, enter the required details, and tap Add. Example: Taylor enters Morning Run, today's date, 5 km and 30 minutes, then saves the activity.
2. Log a Strength activity: Select Strength, enter the activity name and date, add each exercise, and complete all weight, repetition and set-count fields. Example: Taylor records Full Body Session with Squat at 40 kg, 8 reps and 3 sets, plus Bench Press at 25 kg, 10 reps and 3 sets.

3. Add more Strength exercises or set rows: Use Add Exercise for another movement and Add Set for another set configuration. Example: Taylor adds Deadlift as a second exercise and creates a separate lighter warm-up set.
4. Save an activity as a template: Open History and tap the heart icon on a useful saved workout. Example: Taylor hearts the Morning Run record, which then appears under Cardio templates.
5. Use a template: Open Templates, select Cardio or Strength, tap Use Template, review the copied values on the New form, update the date or numbers, and save. Example: Taylor reuses Morning Run, changes the distance from 5 km to 6 km, and logs it as a new activity.
6. Edit an activity: Open History, open the record for editing, correct the fields and save. Example: Taylor changes a run duration from 30 to 32 minutes; the original activity is updated without adding another score reward.
7. Delete an activity: Use the delete action and confirm the permanent removal. Example: Taylor deletes a duplicate Strength record after checking that the correct entry remains.

4.7 Budget Tracker

Function Description

Budget Tracker connects monthly planning, expense recording, Finance scoring and Saving Plans. The user's total monthly budget is divided by the number of days in that month to produce the daily allowance shown in spending views. Each logged expense contributes Finance action evidence for Home scores and Progress Analytics. When the Saving screen is opened on a later day, PULSE compares yesterday's allowance with yesterday's recorded expenses; if money is left, an active saving plan exists and no saving entry was already recorded for that day, it asks whether the unused amount should be added to a saving plan. Eligible budget or spending achievements can also be shared to Social.

Rules

- Budget limits are monthly and are set separately for Food, Transport, Rent, Entertainment and Other. Total monthly budget is the sum of those five limits.
- An expense requires an amount greater than zero and a selected account. Category, account and date are stored; note is optional.
- Daily allowance equals total monthly budget divided by the number of days in the selected month. The calendar compares each day's spending with that allowance.
- The user may browse the current and previous months, but month navigation does not move beyond the current month.
- When the account list is empty on first use, Everyday, Savings, Credit Card and Cash accounts are created. Accounts can then be added, renamed, recoloured or deleted.
- Deleting an account does not rewrite historical expenses that used that account ID. Individual expenses can also be permanently deleted.
- Each newly added expense logs one 25-point Finance expense action.

- The logging streak allows one consecutive empty day; two consecutive missed days end the current streak.

Workflows and Examples

1. Set up budgets before tracking: Open Features, choose Finance, stay on Spending, open Settings, enter monthly limits for the five categories, and save. Example: Taylor sets Food to $500, Transport to $180, Rent to $1,400, Entertainment to $150 and Other to $120, creating a total monthly budget of $2,350.
2. Review or customise accounts: In Finance Settings, rename, recolour, add or delete accounts, then save. Example: Taylor renames Everyday to Main Debit, keeps Cash, adds Joint Account, and removes an unused account.
3. Log an expense: Tap Add expense, enter a positive amount, choose category and account, confirm the date, optionally add a note, and save. Example: Taylor records $18.50 for Food from Main Debit today with the note Lunch.
4. Log a forgotten expense in a past month: Navigate to the earlier month, tap Add expense, adjust the default date if needed, and save. Example: While viewing September, Taylor records a $42 Transport expense dated 28 September.
5. Review spending: Use the donut chart, category bars and calendar to compare spending with the monthly category limits and daily allowance. Example: Taylor sees Food approaching its $500 limit and a red calendar day where spending exceeded the daily allowance.
6. Delete an incorrect expense: Find the transaction, tap its delete action and confirm. Example: Taylor removes a duplicated $18.50 lunch transaction.
7. Build a logging streak: Record at least one expense on tracked days. Example: Taylor logs spending Monday to Friday, misses Saturday, and logs again Sunday; the one-day grace keeps the streak active.

4.8 Saving Plans and Saving Records

Function Description

Saving Plans turns saving goals and dated saving records into live progress for each active plan. It receives the unused-budget opportunity calculated from Budget Tracker, lets the user choose an active plan, and can add that surplus as a saving record for yesterday. All records roll up into the plan’s saved total and remaining amount; crossing a milestone or completing a goal can trigger a celebration and an optional Social achievement share. Saving records remain separate from expense records even though both features are available inside Finance.

Rules

- A saving plan requires a name, a goal amount greater than zero and a due date. Icon and colour can be customised.
- New plans start as Active. Existing plans can be set to Active, Completed or Archived, and completed or archived plans can be restored.

- Only active plans are available when entering saving records. Every entry requires a selected plan and an amount greater than zero.
- Saving records may be added or edited only for today and the previous seven days. Future dates and older dates remain viewable but are not editable.
- A day can contain at most four saving entries. The implementation limits entry count but does not require the entries to use different plans.
- Saving a day replaces all saving entries previously stored for that date with the rows currently shown. Saving with no rows clears that day.
- Progress prompts occur when a plan crosses 25%, 50%, 75% or 90%. Reaching 100% offers completion of the plan.
- The unused-budget offer is checked at most once per day. It requires an active plan, a positive monthly budget and yesterday's spending below the calculated daily allowance.
- Deleting a saving plan permanently removes the plan and all linked saving records.

Workflows and Examples

1. Create a saving plan: Open Finance, switch to Saving, tap Create a saving plan, enter the required goal details and save. Example: Taylor creates Emergency Fund with a $2,000 target, a 31 December due date, a shield icon and a green colour.
2. Add today's saving record: Long-press today in the saving calendar, enter the amount, select an active plan and save. Example: Taylor adds $50 to Emergency Fund.
3. Split one day across several plans: In the record editor, use Add another saving for up to four rows, assign each row and save. Example: Taylor records $50 to Emergency Fund and $20 to Holiday on the same day.
4. View or edit a recent day: Tap a date to view its entries; long-press today or one of the previous seven days to edit. Example: Taylor long-presses yesterday and corrects Emergency Fund from $50 to $45.
5. View an older record: Tap a date older than seven days to read it. Example: Taylor opens a record from last month and can see the saved amounts, but the app does not open the editor.
6. Clear a day's saving records: Open an editable day, remove every row, and save. Example: Taylor removes two entries created on the wrong date; that date then has no saving records.
7. Use the unused-budget offer: When the prompt appears, choose an active plan and add the available amount. Example: Yesterday's allowance was $80 and spending was $55, so Taylor adds the $25 surplus to Emergency Fund.
8. Complete, archive or restore a plan: Edit the plan and change its status, or accept the completion prompt after reaching the goal. Example: Emergency Fund reaches $2,000, Taylor marks it Completed, and later restores it to Active to increase the target.
9. Delete a plan: Choose Delete and confirm that the plan and every linked record will be removed. Example: Taylor deletes a test plan after confirming that its historical entries are no longer needed.

4.9 To-do List

Function Description

To-do List stores personal tasks and links completion to Productivity scoring. The first completion of a task writes one idempotent Productivity action, so repeatedly unticking and reticking the same task cannot create extra score rewards; later score recalculation updates Home, Daily Mission priorities and Progress Analytics. When all current tasks are complete, PULSE can also offer to share the achievement in the Social feed.

Rules

- Task title is required. Description and due date are optional. Priority can be High, Medium or Low.
- Tasks can be viewed as Pending, Completed or All and sorted by due date or priority.
- An incomplete task is overdue only when it has a real due date earlier than today in the device's local time. A task without a due date is not overdue.
- The first completion of a task logs one 25-point Productivity action. A stable reward ID prevents repeated points if the same task is unchecked and completed again.
- Tasks can be edited or permanently deleted after confirmation.
- Completing the final outstanding task can trigger an achievement-sharing prompt.

Workflows and Examples

1. Add a task: Enter the title, optional description and due date, choose priority, and tap Add. Example: Taylor creates Submit project report, adds the description Attach final PDF, chooses Friday as the due date and sets High priority.
2. Complete or reopen a task: Tap its checkbox; tap again to return it to Pending. Example: Taylor completes Submit project report, receives the one-time Productivity action, then reopens it to add a missing attachment without receiving duplicate points later.
3. Filter and sort tasks: Choose Pending, Completed or All, then sort by Due date or Priority. Example: Taylor selects Pending and Priority to place High-priority unfinished tasks first.
4. Edit a task: Tap the task, update its title, description or due date, and save. Example: Taylor changes the report deadline from Friday to Monday.
5. Delete a task: Use the trash icon or long-press delete flow and confirm. Example: Taylor permanently removes a duplicate reminder.
6. Share completion: Complete the last pending task and accept the optional share prompt. Example: After finishing all five tasks, Taylor shares a five-of-five completion achievement with friends.

4.10 Social Feed, Friends and Messaging

Function Description

Social Feed, Friends and Messaging uses the account profile and accepted-friend relationships to control identity, feed visibility and one-to-one conversations. It is also the optional sharing destination for achievements produced by Mood and Journal, Physical Activity, Budget, Saving Plans, To-do List and Daily Missions. Likes, comments, posts and messages do not themselves change domain scores or personal tracker records; they provide encouragement around progress recorded elsewhere.

Rules

- Another user is found by exact email address. A user cannot add or message themselves.
- Each user pair can have only one pending request or one accepted friendship. The recipient can accept or decline; the sender can cancel; either friend can later remove the friendship.
- Feed visibility and direct messaging are limited to the signed-in user and accepted friends.
- A standard daily check-in requires non-empty reflection text and is limited to one per user per local calendar day. Photo is optional.
- Achievement posts are not limited to one per day. Their reflection and photo are optional because the achievement card contains the core information.
- A like is a toggle, so each user has at most one active like on a post. Comments cannot be empty, and only the comment author can delete their comment.
- Only the post owner sees the delete action. Post deletion is permanent, after which the app separately attempts to remove any uploaded image.
- A direct message must contain text, a photo or both. Conversation lists and message threads update in real time.
- The current feed loads the newest 30 posts from at most 30 author IDs including the user. A conversation view loads up to the latest 100 messages.

Workflows and Examples

1. Send and accept a friend request: Open Messages, search the exact email, tap Add friend, and wait for the recipient to accept. Example: Taylor searches jordan@example.com; Jordan accepts the request, and both users then appear in each other’s Friends list.
2. Cancel, decline or remove a connection: Cancel an outgoing request, decline an incoming request, or remove an accepted friend after confirmation. Example: Taylor cancels a request sent to the wrong email before it is accepted.
3. Create a daily check-in post: Use the post composer, write a reflection, optionally attach a photo, and publish. Example: Taylor writes Completed a 5 km run and adds a photo; another daily check-in cannot be created until the next local day.

4. Share an achievement: Accept a share prompt from Journal, Activity, To-do or Finance, choose a template, optionally add text or a photo, and publish. Example: Taylor shares a saving-goal milestone after Emergency Fund reaches 50%; this does not consume the daily check-in slot.
5. Like or comment: Tap the heart to like or unlike, open comments, enter non-empty text, and post. Example: Jordan likes Taylor's running achievement and comments Great consistency; Jordan can later delete that own comment.
6. Delete an own post: Use Delete on a post created by the signed-in user and confirm. Example: Taylor removes a post containing the wrong photo; friends can no longer see it.
7. Start a direct message: Tap an accepted friend, enter text or attach a photo, and send. Example: Taylor opens Jordan's conversation and sends Are you free for a walk? with an optional image.
8. Read conversations: Open a thread to load the latest messages and mark it read. Example: Taylor opens a thread with an unread indicator; returning to the list clears that indicator.

4.11 Progress Analytics

Function Description

Progress Analytics is the read-only reporting destination for weekly score snapshots created by the scoring system. Those snapshots ultimately combine the Entry Quiz baseline with reflections and actions from journals, workouts, tasks, expenses, domain activities and Daily Missions. Selecting Overall or a domain changes which stored series is analysed, while new activity elsewhere in PULSE affects the next recalculated snapshot rather than allowing users to type chart values directly.

Rules

- The app reads up to the latest 12 weekly snapshots.
- The available time ranges are four, eight and twelve weeks.
- The user can analyse the overall Balanced Life Score or any one of the five domains.
- Growth percentages and average weekly change require at least two saved weekly data points.
- Charts and insights are calculated from saved scores; users do not manually enter trend values.

Workflows and Examples

1. Review overall progress: Open Profile and Progress Analytics, choose Overall, then select a time range. Example: Taylor selects 8 Weeks and sees the Balanced Life Score trend and percentage growth from the first to the latest week.

2. Investigate a domain: Select one domain to replace the overall line with that domain's history and insight. Example: Taylor selects Finance and sees whether budget and saving activity have improved the Finance score over the chosen period.
3. Compare different periods: Switch between 4, 8 and 12 Weeks. Example: Taylor uses 4 Weeks for recent change, then 12 Weeks to check whether the improvement is sustained.

4.12 Guided Tutorials

Function Description

Guided Tutorials connects first-use guidance to the real routes and semantic interface targets used across Home, Finance, Saving Plans, activity trackers, Social and Profile. Tutorial progress is stored for the signed-in account and can be replayed from Profile. Explanation-only targets block normal taps; an intentional interaction is exposed through Try it now, and route synchronisation plus automatic scrolling keeps the active screen, target and tutorial step aligned without changing normal navigation or scrolling when the tutorial is inactive.

Rules

- Each tutorial has a stored version and appears when that version has not yet been completed for the user.
- Every tutorial step belongs to a specific app screen. The tutorial activates the correct route before displaying the step.
- Before drawing a spotlight, the app scrolls the target into view and measures the live interface element. If a valid target cannot be measured, an accessible fallback card is used.
- Explanation-only highlights block taps on the underlying control. Intentionally interactive steps use an explicit Try it now action.
- Users can replay an individual tutorial or restart all tutorials from Profile settings.

Workflows and Examples

1. Follow a first-use tutorial: Read the explanation and use Next or Back. Example: During the Budget tutorial, Next opens the correct Finance view and highlights the relevant budget control before showing the next explanation.
2. Try an instructed action: Use Try it now only when the tutorial offers it. Example: On Record an expense, Taylor taps Try it now and the tutorial deliberately opens the expense form.
3. Replay a tutorial: Open Profile, Settings, Help & tutorials, and choose a tutorial. Example: Taylor replays Saving Plans to review how recent saving records are edited.
4. Restart all tutorials: Choose Restart all tutorials in Help & tutorials. Example: After handing the phone to a new tester, Taylor resets the tutorial progress so the first-use guidance appears again.

5 Technology Stack

5.1 Core Application Platform

PULSE uses one shared JavaScript codebase to deliver the Android, iOS and web versions of the application.

- JavaScript — the primary language for application logic, screen behaviour, scoring and data services. Official reference: MDN JavaScript.
- React 19.2.0 — the component-based user interface foundation used to build reusable screens and controls. Official website: React.
- React Native 0.83.10 — renders the shared React components as native Android and iOS interface elements. Official website: React Native.
- Expo SDK 55.0.28 — supplies the development runtime, native-module integration, Metro workflow and web export used by the project. Official website: Expo.

5.2 Backend and Data Storage

- Firebase JavaScript SDK 11.10.0 — connects the application to Firebase Authentication, Cloud Firestore and Cloud Storage. Official website: Firebase.
- Firebase Authentication — manages email-and-password accounts, verification, password reset and persistent signed-in sessions. Management website: Firebase Console.
- Cloud Firestore — stores profiles, consent, scores, missions, journals, workouts, budgets, saving plans, tasks, posts, friends and messages. Management website: Firebase Console.
- Cloud Storage — stores uploaded profile and social-post media. Management website: Firebase Console.
- AsyncStorage 2.2.0 — stores selected device-side state, including persistent native authentication support and local acknowledgement state. Official project: AsyncStorage.
- Expo SecureStore 55.0.16 — provides protected device storage where sensitive local values are required. Official documentation: SecureStore.

5.3 Mobile and Interface Libraries

- Expo Image Picker 55.0.22 — accesses the camera or media library for profile and post images. Official documentation: Image Picker.
- Expo Font 55.0.8 and @expo-google-fonts/manrope 0.4.2 — load and apply the Manrope typeface. Official documentation: Expo Fonts.
- @expo/vector-icons 15.0.2 — supplies interface icons used throughout the app. Official documentation: Expo Vector Icons.
- React Native DateTimePicker 8.6.0 and Picker 2.11.4 — provide native date, time and selection controls. Official projects: DateTimePicker and Picker.

- React Native SVG 15.15.3 and react-native-qrcode-svg 6.3.21 — render vector graphics and QR codes. Official projects: React Native SVG and QR Code SVG.
- React Native Safe Area Context 5.6.2 and React Native Screens 4.23.0 — support safe device insets and native screen containers. Official projects: Safe Area Context and React Native Screens.

5.4 Application Architecture

- Presentation layer — React Native pages and reusable components render the interface and collect user actions.
- Application layer — the root app state coordinates authentication, onboarding, consent, tabs, stacked feature screens and tutorial routing.
- Service layer — feature-specific Firebase modules isolate account, scoring, finance, savings, social and messaging operations from the screen code.
- Domain-logic layer — the scoring and Daily Mission engines are independent JavaScript modules, allowing calculations and selection rules to be tested without rendering the interface or contacting Firebase.

5.5 Testing and Quality Assurance

- Jest 29.7.0 — runs the automated JavaScript test suite. Official website: Jest.
- jest-expo 55.0.7 — supplies the Expo-compatible Jest preset and mocks. Official documentation: Expo unit testing.
- React Native Testing Library 13.2.0 — tests components through user-visible behaviour and accessibility queries. Official website: React Native Testing Library.
- React Test Renderer 19.2.0 — supports React component rendering in the test environment. Official reference: React Test Renderer.
- Quality activities — automated unit and component tests, simulation tests, Android emulator/device checks and manual user testing are used together.

5.6 Development, Handover and Android Emulator Software

These are the main applications and websites a Windows stakeholder needs to open, run and maintain the handed-over project.

- IntelliJ IDEA — the main integrated development environment for opening, reading and editing the project. Official download page: IntelliJ IDEA.
- Git for Windows — provides local version control and the Git command-line tools used by IntelliJ IDEA and GitHub Desktop. Official download page: Git for Windows.
- GitHub and GitHub Desktop — GitHub hosts the handed-over repository; GitHub Desktop provides a simpler Windows interface for cloning, pulling and reviewing changes. Websites: GitHub and GitHub Desktop download.

- Node.js LTS and npm — run the Expo development tools and install the project dependencies. Official download page: Node.js.
- Android Studio — installs the Android SDK and provides Device Manager for creating and running the Android Virtual Device emulator. Official download page: Android Studio.
- Metro — the JavaScript development bundler started through Expo when the project runs locally. Official documentation: Metro.
- Expo Application Services (EAS) Build — the hosted Expo service used to create installable Android and iOS application binaries. Official documentation: EAS Build.